



Intelligent analysis of dynamic characteristics of steel-concrete hybrid wind turbine tower based on adaptive vibration mode

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ABSTRACT

The steel-concrete wind turbine hybrid tower addresses the high flexibility and transportation difficulties associated with traditional steel towers, making it increasingly popular in onshore low-wind-speed wind farms. Accurate analysis of dynamic characteristics plays an important role in the design of wind turbine tower structures. The vibration governing equation for the hybrid tower was derived using the principle of virtual displacement and the generalized coordinates method, considering the mass eccentricity and moment of inertia of the rotor-nacelle assembly, variable cross-sections of the tower, and prestress. An adaptive vibration mode for the hybrid tower was constructed, and an intelligent optimization algorithm was employed to solve the natural frequencies. A refined finite element model was developed using ABAQUS software to predict the natural frequencies and vibration modes of the hybrid tower, validating the reliability of the intelligent analysis approach. Moreover, the impact of different simplified rotor-nacelle assembly models on the natural frequencies of the hybrid tower was explored, revealing that models neglecting moments of inertia overestimate structural natural frequencies, particularly for higher modes. Finally, the feature importance of various geometric dimensions of the tower was assessed employing a machine learning model, and parametric analysis of the geometric dimensions was conducted. The results show that tower height, proportion of concrete tower segment height, and bottom diameter of concrete tower segment are critical factors influencing the natural frequencies.

1. Introduction

With the continuous development of the global economy and the increasing consumption of energy, traditional energy sources such as coal, oil, and natural gas are facing supply shortages. Consequently, the development of new renewable energy sources has become a critical strategic demand for many countries. Wind power, as a clean and efficient renewable energy source, can effectively address energy consumption concerns. At present, many countries around the world are vigorously developing wind power, with installed capacity increasing year by year [1–3].

To achieve higher power output, wind turbines are being developed towards larger scales. The increasing length of wind turbine blades and the height of support structures pose new and significant challenges to the mechanical performance and economic feasibility of wind turbine towers [4]. For hub heights exceeding 100 m, traditional steel towers

induce resonance issues and are difficult to manufacture and transport, severely limiting the advancement of wind power. Steel-concrete wind turbine hybrid tower emerges as an innovative solution, featuring a conical concrete lower section, a concrete transition segment in the middle, and a traditional steel upper section. Vertical prestress is applied to the concrete using external prestressed steel strands, as shown in Fig. 1. This hybrid tower combines the advantages of both concrete and steel towers, offering high load-bearing capacity, excellent stability, and ease of transportation and construction, which makes it particularly suitable for the construction of tall wind farms [5–8]. In recent years, hybrid towers have been widely applied to wind turbines with hub heights exceeding 100 m in low-wind speed regions [9].

Wind turbines will experience significant vibrations under wind loads and their own vibration loads. When the operational frequency of the wind turbine approaches the natural frequency of the overall structure, resonance occurs, which can affect the performance of the

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wind turbine and even lead to structural failure. To avoid resonance between the tower and the wind turbine, the first-order frequency of the tower should steer clear of the $\pm 10\%$ range of the wind turbine rotational frequency f_{1p} and the blade passing frequency f_{3p} . Wind turbine towers are generally designed with three strategies: the stiff-stiff strategy with natural frequencies greater than $1.1 f_{3p}$, the soft-stiff strategy with natural frequencies between $1.1 f_{1p}$ and $0.9 f_{3p}$, and the soft-soft strategy with natural frequencies less than $0.9 f_{1p}$ [10,11]. Therefore, precise analysis of the dynamic characteristics of wind turbine tower structures is crucial to ensure their safe and effective operation.

Currently, research on the natural frequencies of wind turbine towers primarily involves analytical calculations and numerical simulations. In terms of numerical simulations, Tian et al. [12] treated the wind turbine rotor, nacelle, and flange as equivalent mass points, and utilized shell elements to construct a finite element (FE) model of the tower structure for natural frequency calculations, validated through experimental verification of computational accuracy. Alkhoury et al. [13] used point masses and beam elements to construct a nacelle model, studying the characteristic modes and frequencies of the wind turbine structure. Cao et al. [14] adopted shell elements to develop a detailed FE model of a monopile-tower-blade system, considering soil reactions, wind-wave combinations. Shi et al. [15] investigated the impact of four different simplified rotor-nacelle assembly (RNA) models on the characteristic frequencies and seismic responses of offshore wind turbine structures, finding that simplified RNA models will affect higher-order modes. Lana et al. [17] established a solid model of a prestressed concrete tower to calculate the first five natural frequencies and modes, and optimized the shape and dimensions of the hybrid tower using a genetic algorithm. Kenna et al. [16] created a detailed FE model of a prestressed concrete tower using two-dimensional continuum shell elements and one-dimensional beam elements, exploring the impact of prestress design and concrete compressive strength on tower stiffness. Although detailed FE models can accurately determine the natural frequencies of wind turbine towers, the modeling workload is substantial, making it unsuitable for simplified and rapid calculations during the early design stages of the towers.

Analytical calculation of tower natural frequencies offers a clear

mechanism and serves as a supplementary method to validate the accuracy of numerical results. Analytical solutions of natural frequencies are useful for preliminary tower design, avoiding the need for repetitive numerical modeling and enhancing design efficiency. Vugst [18] and Tempel [19] treated tower structures as cantilever beams, with nacelles and wind rotors equivalent to concentrated masses at the top, deriving analytical formulas for natural frequencies based on structural dynamics principles. Yang et al. [20] employed the Rayleigh method and the Lagrange equation to calculate natural frequencies of single pile offshore wind turbine pure steel tower, demonstrating that tower height, base diameter, wall thickness, pile diameter, and length are significant factors influencing natural frequencies in offshore wind turbine structures. Lei et al. [21] considered the effects of wind turbine mass and eccentricity, deriving analytical methods for calculating natural frequencies for steel tower using energy methods. Huang et al. [22] obtained the natural frequencies of steel-concrete hybrid towers by solving free vibration equations, and analyzed their correlation with geometric dimensions. Zhang et al. [23] used flexibility matrices to calculate natural frequencies of precast concrete towers and studied the impact of top concentrated mass as well as precast segmental thickness on structural dynamic characteristics. Li et al. [24] employed the Rayleigh-Ritz method to establish frequency equations for steel-concrete hybrid towers, systematically considering turbine mass, rotational inertia, elastic foundation stiffness, and prestress effects on structural natural frequencies. However, top wind turbines have large moment of inertia and their center of mass deviates from the tower axis. Scholars often simplify top units into concentrated mass point, neglecting rotational inertia, mass eccentricity, and asymmetric effects in two directions on overall structural natural frequencies. Additionally, the accuracy of energy method solutions for tower natural frequencies depends on the selection of mode shapes, and conventional mode shape constructions often struggle to meet engineering accuracy requirements.

The structural configuration and mechanical features of a steel-concrete hybrid tower are relatively complex, with abrupt changes in tower cross-sections at the transition segment. As wind turbines trend towards larger scales, the impact of top unit rotational inertia and mass eccentricity on structural natural frequencies becomes significant and

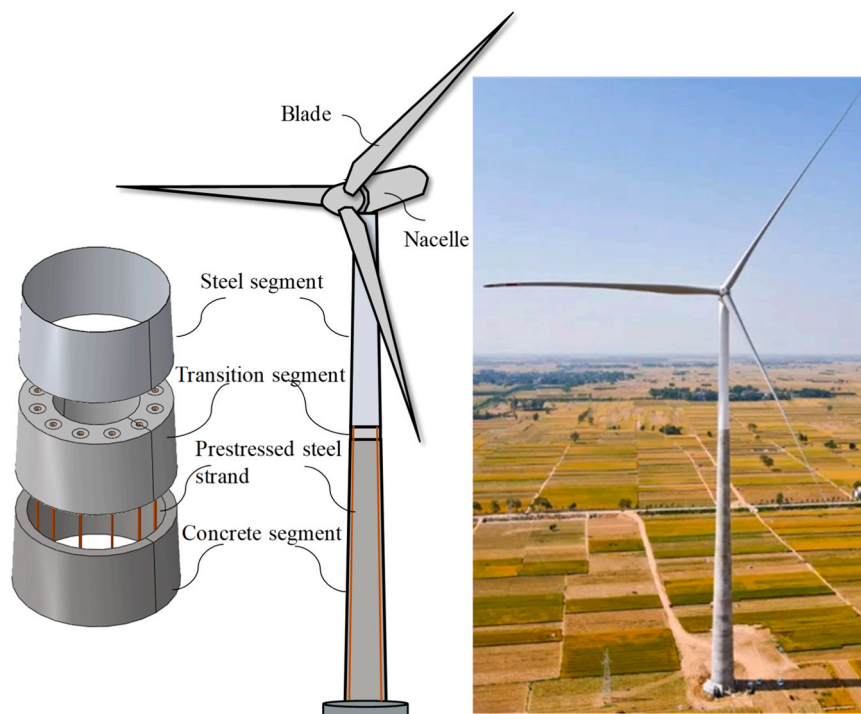


Fig. 1. Steel-concrete hybrid wind turbine tower.

cannot be ignored. After considering numerous influencing factors, the challenge lies in selecting appropriate mode shapes for solving the natural frequencies of hybrid tower structures. Currently, research on the vibration characteristics of steel-concrete hybrid towers is in its initial stages, with theoretical studies lacking sufficient depth to meet engineering practical needs.

Considering the influence of wind turbine mass eccentricity, moment inertia, and prestress in concrete tower segments, the differential equation governing the free vibration of hybrid towers is derived based on the principle of virtual displacement and the adaptive mode shapes, and natural frequencies are solved using a combination of the energy method and intelligent optimization algorithm. A refined FE model of hybrid towers in ABAQUS was developed, validating the reliability of proposed analytical solutions. The feature importance of the geometric dimensions of the hybrid tower was analyzed based on machine learning model, and parameter analysis was conducted on the selected features, aiming to provide design guidelines for the engineering application of hybrid towers.

2. Development of governing equation

2.1. Mechanical model

The steel-concrete hybrid tower primarily is composed of the top RNA, the concrete tower segment, the transition segment, and the steel tower segment. The RNA comprises the nacelle and rotor, which can be modeled as rigid points N and R with concentrated mass and rotational inertia, respectively. The hybrid tower can be simplified as a Euler-Bernoulli cantilever beam with eccentrically concentrated mass and rotational inertia at the free end. A spatial Cartesian coordinate system is established in Fig. 2, with the origin at the center of the tower base. The

coordinates u and v represent the lateral directions of the tower, while x represents the longitudinal coordinate. The heights of the concrete tower segment, transition segment, and steel tower segment are denoted as h_c , h_t , and h_s , respectively, with the total height of the tower being H . At any height x of the tower, the cross-section is annular, with diameter $D(x)$ and wall thickness $t(x)$. The intersection point W of the line connecting the nacelle and rotor center of gravity with the tower axis is at a distance a from the top of the tower. The masses of the nacelle and rotor are m_N and m_R , respectively. The rotational inertias of the nacelle about the u and v axes are J_{Nu} and J_{Nv} , while those of the rotor are J_{Ru} and J_{Rv} . The centroids of the nacelle and rotor are eccentrically located only in the u direction, with eccentric distances Δ_N and Δ_R , respectively.

2.2. Virtual work analysis

The governing equation for steel-concrete hybrid tower is established based on the virtual displacement principle. Below is a detailed derivation of the virtual work performed by the prestress, tower, and top RNA components corresponding to their respective virtual displacements.

(a) Virtual Work Done by prestress

The prestressed steel strands at the top of the transition section in the hybrid tower are anchored, while their bottoms are anchored within cavities inside the foundation. Therefore, prestress is considered to exert an axial force P at the top of the transition section. $u(x,t)$ and $\delta u(x,t)$ respectively represent the lateral displacement and virtual displacement along the u -direction at any height x of the hybrid tower varying continuously with time t . Similarly, $v(x,t)$ and $\delta v(x,t)$ respectively denote the lateral displacement and virtual lateral displacement along the v -

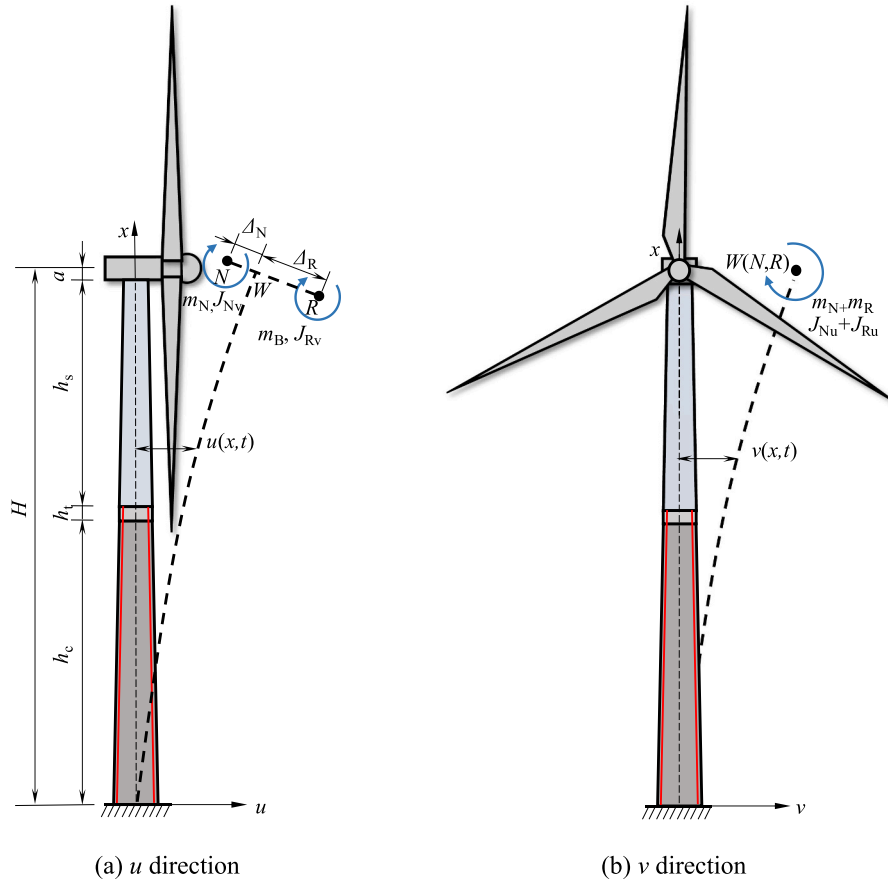


Fig. 2. Simplified analysis model of the steel-concrete hybrid tower.

direction at any height x of the hybrid tower varying continuously with time t . The vertical displacement $w(x, t)$ and virtual displacement $\delta w(x, t)$ at any height x of the hybrid tower varying continuously with time t can be respectively expressed as Eqs. (1) and (2):

$$w(x, t) = \frac{1}{2} \int_0^x \left[\left(\frac{\partial u(z, t)}{\partial z} \right)^2 + \left(\frac{\partial v(z, t)}{\partial z} \right)^2 \right] dz \quad (1)$$

$$\delta w(x, t) = \int_0^x [u'(z, t) \delta u'(z, t) + v'(z, t) \delta v'(z, t)] dz \quad (2)$$

During the lateral deformation process of the tower, the vertical position of the transition section changes, which affects the work done by the prestress during this process. Substituting the height of the prestress application point ($h_c + h_t$) into Eq. (2) to determine the vertical virtual displacement and displacement δw ($h_c + h_t, t$). Therefore, the virtual work done by the prestress P on the vertical virtual displacement δW_p is

$$\delta W_p = P \delta w(h_c + h_t, t) = P \int_0^{h_c + h_t} [u'(z, t) \delta u'(z, t) + v'(z, t) \delta v'(z, t)] dz \quad (3)$$

(b) Virtual Work Done by tower

The hybrid tower is typically a variable cross-section beam, where the bending stiffness $EI(x)$ and mass per unit length $m(x)$ vary arbitrarily along the height x of the tower. E_i and ρ_i represent the elastic modulus and mass density of each segment, respectively. Then, $m(x)$ and $EI(x)$ can be respectively expressed as:

$$m(x) = \pi \rho_i [D(x) - t(x)] t(x) \quad (4)$$

$$EI(x) = E_i \frac{\pi}{64} \{ [D(x)]^4 - [D(x) - 2t(x)]^4 \} \quad (5)$$

The virtual work done by the tower can be divided into three parts: one part is the virtual work done by the tower self-weight in its corresponding vertical virtual displacement δW_G^T ; another part is the virtual work done by the inertial force of the distributed mass of the tower in its corresponding lateral virtual displacement δW_I^T ; and the third part is the virtual work done by the internal bending moment of the tower in its corresponding virtual curvature δW_M^T .

The virtual work δW_G^T done by the tower self-weight on the vertical virtual displacement is

$$\begin{aligned} \delta W_G^T &= \int_0^H m(x) g \delta w(x, t) dx \\ &= \int_0^H \left[m(x) g \left(\int_0^x [u'(z, t) \delta u'(z, t) + v'(z, t) \delta v'(z, t)] dz \right) \right] dx \end{aligned} \quad (6)$$

The virtual work done by the lateral inertial forces $f_{lu}^T(x, t)$ and $f_{lv}^T(x, t)$ on their corresponding lateral virtual displacements is respectively represented as:

$$\delta W_{lu}^T = \int_0^H f_{lu}^T(x, t) \delta u(x, t) dx = - \int_0^H m(x) \ddot{u}(x, t) \delta u(x, t) dx \quad (7)$$

$$\delta W_{lv}^T = \int_0^H f_{lv}^T(x, t) \delta v(x, t) dx = - \int_0^H m(x) \ddot{v}(x, t) \delta v(x, t) dx \quad (8)$$

The total virtual work done by the inertia forces of the tower on corresponding virtual displacements is

$$\delta W_I^T = \delta W_{lu}^T + \delta W_{lv}^T \quad (9)$$

The virtual work done by the lateral inertia moments $M_u(x, t)$ and $M_v(x, t)$ on their corresponding virtual curvatures is respectively represented as:

$$\delta W_{Mu}^T = \int_0^H M_u(x, t) \delta \kappa_u(x, t) dx = \int_0^H -EI(x) u''(x, t) \delta u''(x, t) dx \quad (10)$$

$$\delta W_{Mv}^T = \int_0^H M_v(x, t) \delta \kappa_v(x, t) dx = \int_0^H -EI(x) v''(x, t) \delta v''(x, t) dx \quad (11)$$

The total virtual work done by the tower's bending moment on the virtual curvature is

$$\delta W_M^T = \delta W_{Mu}^T + \delta W_{Mv}^T \quad (12)$$

(c) Virtual Work Done by RNA

Due to the eccentricity of the nacelle and rotor in the u -direction, there is an additional bending moment caused by the eccentricity of the mass at the tower top. To determine the work done by the top RNA, the gravity of the nacelle and blades is shifted to point W on the tower axis. The work done by the top RNA can be divided into four parts: one part is the virtual work δW_G^{WT} done by the self-weight of the RNA on its corresponding vertical virtual displacement; a second part is the virtual work δW_I^{WT} done by the inertial force generated by the mass of the RNA on its corresponding lateral virtual displacement; a third part is the virtual work δW_J^{WT} done by the rotational inertia of the RNA on its corresponding virtual angular displacement; and the fourth part is the virtual work δW_E^{WT} done by the additional bending moment caused by the mass eccentricity of the RNA on its virtual angular displacement.

The displacements u_w , v_w and w_w at point W on the tower axis are respectively represented as

$$u_w = u(H, t) + au'(H, t) \quad (13)$$

$$v_w = v(H, t) + av'(H, t) \quad (14)$$

$$w_w = \frac{1}{2} \int_0^H \left[\left(\frac{\partial u(z, t)}{\partial z} \right)^2 + \left(\frac{\partial v(z, t)}{\partial z} \right)^2 \right] dz + \frac{1}{2} a[u'(H, t)]^2 + \frac{1}{2} a[v'(H, t)]^2 \quad (15)$$

The virtual displacements δu_w , δv_w and δw_w at point W are respectively represented as

$$\delta u_w = \delta u(H, t) + a \delta u'(H, t) \quad (16)$$

$$\delta v_w = \delta v(H, t) + a \delta v'(H, t) \quad (17)$$

$$\begin{aligned} \delta w_w &= \int_0^H [u'(z, t) \delta u'(z, t) + v'(z, t) \delta v'(z, t)] dz \\ &\quad + au'(H, t) \delta u'(H, t) + av'(H, t) \delta v'(H, t) \end{aligned} \quad (18)$$

The linear velocities at point W along the u and v directions are respectively represented as

$$\dot{u}_w = \dot{u}(H, t) + a \dot{u}'(H, t) \quad (19)$$

$$\dot{v}_w = \dot{v}(H, t) + a \dot{v}'(H, t) \quad (20)$$

The linear accelerations at point W along the u and v directions are respectively represented as

$$\ddot{u}_w = \ddot{u}(H, t) + a \ddot{u}'(H, t) \quad (21)$$

$$\ddot{v}_w = \ddot{v}(H, t) + a\ddot{v}'(H, t) \quad (22)$$

The total mass of the top RNA is $m_w (m_N + m_R)$. The virtual work done by the self-weight of the top RNA on the vertical virtual displacement is

$$\delta W_G^{WT} = m_w g \delta u_w \quad (23)$$

The virtual work done by the lateral inertial forces $f_{lu}^M(x, t)$ and $f_{lv}^M(x, t)$ of the RNA on their corresponding lateral virtual displacements are respectively represented as:

$$\delta W_{lu}^{WT} = -m_w \ddot{u}_w \delta u_w \quad (24)$$

$$\delta W_{lv}^{WT} = -m_w \ddot{v}_w \delta v_w \quad (25)$$

The total virtual work done by the inertia forces of the RNA on corresponding virtual displacements is

$$\delta W_I^{WT} = \delta W_{lu}^{WT} + \delta W_{lv}^{WT} \quad (26)$$

The virtual work done by the rotational inertia moments of the top RNA on their corresponding virtual angular displacements are respectively represented as:

$$\delta W_{Ju}^{WT} = -(J_{Nu} + m_N \Delta_N^2) \ddot{u}'_w \delta u'_w - (J_{Ru} + m_R \Delta_R^2) \ddot{u}'_w \delta u'_w \quad (27)$$

$$\delta W_{Jv}^{WT} = -(J_{Nv} + J_{Rv}) \ddot{v}'_w \delta v'_w \quad (28)$$

The total virtual work done by the rotational inertia moments of the top RNA is expressed as:

$$\delta W_J^{WT} = \delta W_{Ju}^{WT} + \delta W_{Jv}^{WT} \quad (29)$$

The virtual work done by the additional moment caused by the mass eccentricity of the top wind turbine on its virtual angular displacement is

$$\delta W_E^{WT} = (m_R g \Delta_R - m_N g \Delta_N) \delta u'_w \quad (30)$$

According to the principle of virtual work, the total virtual work done by the whole wind turbine tower structure δW_E is 0,

$$\delta W_E^p + \delta W_G^T + \delta W_I^T + \delta W_M^T + \delta W_G^{WT} + \delta W_J^{WT} + \delta W_I^{WT} + \delta W_E^{WT} = 0 \quad (31)$$

2.3. Development of governing equation

According to principles of structural dynamics, assuming a shape function $\varphi(x)$ for a specific mode of vibration of the hybrid tower, where the motion amplitude relative to the tower base along the u -direction is generalized coordinate $T_u(t)$, and the motion amplitude relative to the tower base along the v -direction is generalized coordinate $T_v(t)$, then the displacements of the hybrid tower at height x at time t along the u -direction and v -direction are respectively

$$u(x) = \varphi(x) T_u(t) \quad (32)$$

$$v(x) = \varphi(x) T_v(t) \quad (33)$$

By substituting Eqs. (32–33) into Eq. (31), the governing equation can be obtained:

$$\left(M_u \ddot{T}_u + K_u T_u + C_u \right) \delta T_u + \left(M_v \ddot{T}_v + K_v T_v \right) \delta T_v = 0 \quad (34)$$

where:

$$M_u = (J_{Nu} + m_N \Delta_N^2 + J_{Ru} + m_R \Delta_R^2) [\varphi'(H) + a\varphi''(H)]^2 + m_w [\varphi(H) + a\varphi'(H)]^2 + \int_0^H m(x) [\varphi(x)]^2 dx \quad (35)$$

$$K_u = K_v = -P \int_0^{h_1+h_2} [\varphi'(z)]^2 dz - \int_0^H \left[m(x) g \left(\int_0^x [\varphi'(z)]^2 dz \right) \right] dx + \int_0^H EI(x) [\varphi''(x)]^2 dx - m_w g \left\{ \int_0^H [\varphi'(x)]^2 dx + a [\varphi'(H)]^2 \right\} \quad (36)$$

$$C_u = -(m_R g \Delta_R - m_N g \Delta_N) [\varphi'(H) + a\varphi''(H)] \quad (37)$$

$$M_v = (J_{Nv} + J_{Rv}) \left\{ [\varphi'(H)]^2 + 2a\varphi''(H)\varphi'(H) + a^2[\varphi''(H)]^2 \right\} + m_w [\varphi(H) + a\varphi'(H)]^2 + \int_0^H m(x) [\varphi(x)]^2 dx \quad (38)$$

If for any variation δT_u and δT_v , Eq. (34) holds true, then

$$M_u \ddot{T}_u + K_u T_u + C_u = 0 \quad (39)$$

$$M_v \ddot{T}_v + K_v T_v = 0 \quad (40)$$

Eqs. (39) and (40) respectively represent the motion governing equations of the tower along the u and v directions. By substituting the shape function $\varphi_i(x)$ for the i -th mode of tower vibration, the generalized masses M_i^u and M_i^v , and the generalized stiffnesses K_i^u and K_i^v along the u and v directions are calculated. Therefore, according to Rayleigh energy method, the natural frequencies of the hybrid tower along the u and v directions are respectively represented as:

$$f_i^u = \sqrt{\frac{K_i^u}{M_i^u}} \quad (41)$$

$$f_i^v = \sqrt{\frac{K_i^v}{M_i^v}} \quad (42)$$

Eqs. (41) and (42) respectively yield the natural frequencies of tower vibration along the u and v directions. This is because the rotational inertia and eccentric mass of the wind turbine atop the tower differ around the u and v axes, resulting in different natural frequencies of vibration along each direction for each vibration mode.

3. Solution of governing equation

3.1. Adaptive vibration mode function

The accuracy of the natural frequency calculation depends on the similarity between the assumed vibration mode function and the actual vibration mode function. The shape function for the i -th vibration mode of a cantilever beam with constant cross-section is expressed as

$$\psi_i(x) = \cos a_i x - \cosh a_i x - \frac{\cos a_i H + \cosh a_i H}{\sin a_i H + \sinh a_i H} (\sin a_i x - \sinh a_i x) \quad (43)$$

where a_i is determined by the transcendental equation $\cosh a_i H = -(1/\cosh a_i H)$.

Eq. (43) is the exact solution for the vibration mode function of a homogeneous cantilever beam with constant cross-section. However, since the hybrid tower is a cantilever beam with variable cross-section and material properties, and is influenced by the top wind turbine, using Eq. (43) alone may result in significant errors. By superimposing the various vibration modes of a cantilever beam with equal cross-section, a new shape function for the i -th vibration mode can be constructed to closely approximate the actual vibration mode function of a hybrid tower.

$$\phi_i(x) = \sum_{j=1}^n q_j \psi_j(x) \quad (44)$$

where q_j represents the undetermined coefficient of the j -th vibration mode function $\psi_j(x)$.

By using a filtering method, higher-order vibration mode functions can be obtained [25]. If the contribution of the first vibration mode to the assumed vibration mode function is zero, then the second vibration mode will dominate. To calculate the second mode shape, it is sufficient to assume $q_1 = 0$, thereby constructing a vibration mode function that

does not include the first vibration mode component of a constant cross-section cantilever beam. Similarly, setting both q_1 and q_2 to zero will yield the third vibration mode.

3.2. Equation solving

According to the Rayleigh method, the frequency obtained by the actual vibration mode is the lowest. Therefore, when selecting the approximate results obtained using this method, the one with the lowest frequency is always the best approximation [25]. This can be framed as an optimization problem aimed at minimizing a function subject to specified constraints, represented by the following mathematical model:

$$\begin{aligned} &\text{find : } \mathbf{q} = \{q_1, q_2, \dots, q_n\} \\ &\text{minimize : } f_i(\mathbf{x}) \\ &\text{subject to : } -1 \leq q_j \leq 1 \quad (j = 1, 2, \dots, n) \end{aligned} \quad (45)$$

According to Eq. (45), the undetermined coefficients q_j can be obtained, thereby simultaneously determining the vibration modes and natural frequencies of each order. An intelligent optimization algorithm is employed to solve Eq. (45). The genetic algorithm is a global search optimization algorithm that simulates natural selection and genetic mechanisms, enabling it to search for the optimal solution globally. This makes it particularly suitable for complex, nonlinear, and non-smooth objective function optimization problems, avoiding local optima [26, 27]. Therefore, the genetic algorithm is used to solve Eq. (45) in this study. Fig. 3 illustrates the flowchart for calculating the dynamic characteristics of hybrid tower. The main parameters of algorithms were set as follows: the number of population size was 50, the number of iterations $i_{\max}$ was 100, the crossover probability was set to 0.8, and the mutation probability was set to 0.01.

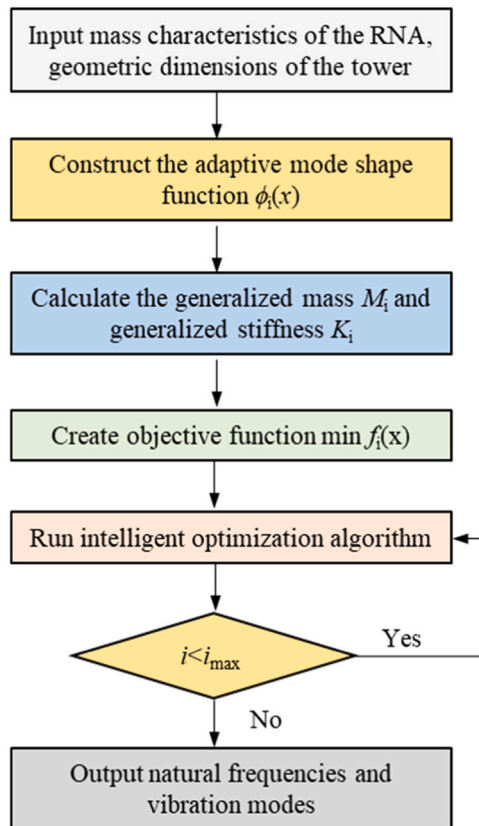


Fig. 3. The flowchart of the dynamic characteristics of hybrid tower.

3.3. Case

In this study, a 5.0 MW hybrid tower with a height of 160 m is selected as a case study. The mass of this wind turbine is 234 t. The rated wind speed and cut-out wind speed are 11 m/s and 25 m/s, respectively. The concrete tower is made of C80 concrete, with an elastic modulus of 36,000 MPa and a density of 2500 kg/m³. The steel grade of steel tower is Q355, with an elastic modulus of 2.06×10^5 MPa. To account for the contributions of welding, bolts, coating, and flanges to the mass of the steel tower, the equivalent density of the steel tower is taken as 8500 kg/m³ to consider the effect of paint and connections [15,24]. A prestress value of 80 MN is applied to the top of the transition segment. The geometric parameters of the tower and the mass characteristics of the top wind turbine are listed in Tables 1 and 2, respectively.

The curves of the first three natural frequencies of the hybrid tower with the number of vibration mode components are obtained by programming in Python, as shown in Fig. 4. As the number of vibration mode components increases, the first three natural frequencies gradually decrease and converge. When the constructed adaptive mode shape function consists of 8 vibration mode components superimposed, the first three natural frequencies converge to satisfactory results. Due to the significant increase in the moment of inertia of wind turbines around the u -axis compared to the v -axis, the generalized mass along the v -direction of the tower is greater than along the u -direction. Therefore, the natural frequencies of vibration along the u -direction are slightly higher than those along the v -direction. Additionally, the differences in the second and third natural frequencies along both directions are more pronounced than in the first frequency, indicating that moments of inertia and eccentric mass primarily influence the higher-order natural frequencies of the tower vibration.

4. Finite element analysis

4.1. Modeling

A FE analysis model was established for steel-concrete hybrid tower by ABAQUS software. The model comprises components such as concrete tower segment, transition segment, prestressed steel strands, and steel tower segment. C3D8R solid elements were adopted for concrete tower segment, transition segment, and steel tower segment, while B31 beam element was adopted for prestressed steel strands. Both steel and concrete materials employed elastic constitutive laws as detailed in Section 2.3 for their modulus and density values. Tie constraints were utilized to simulate the connections between tower segments. The upper nodes of the prestressed steel strands were constrained to the anchor faces at the top of the transition segment using MPC (multi-point constraint) constraints, while the lower nodes are restricted in all translational degrees of freedom to simulate hinged boundary conditions. To apply the prestress, initial temperatures were defined for the strands, with subsequent definitions of final temperatures in the loading steps to account for tendon shrinkage. The linear expansion coefficient for the prestressed steel strands was set to $1e-5$. Reference points were established at the center of the nacelle and rotor, respectively, and

Table 1
The geometric parameters of the hybrid tower.

Geometric parameters	Values /m
Top and bottom diameters of the concrete tower segment	10 , 4.8
Top and bottom diameters of the transition segment	4.8 , 4.6
Top and bottom diameters of the steel tower segment	4.6 , 4
Top and bottom thicknesses of the concrete tower segment	0.23 , 0.34
Top and bottom thicknesses of the transition segment	1.6 , 1.2
Top and bottom thicknesses of the steel tower segment	0.018 , 0.012
Height of the concrete tower segment	105
Height of the transition segment	2.5
Height of the steel tower segment	50

Table 2

The mass characteristics of the RNA.

Component	Mass/t	Centroid/m	Moment of inertia	
			$J_u/\text{kg}\cdot\text{m}^2$	$J_v/\text{kg}\cdot\text{m}^2$
Nacelle	125	(1.082,0,2.5)	1.58×10^5	1.08×10^6
Blade	121	(−5.825,0,2.5)	9.99×10^7	5.30×10^7
RNA	246	(−2.315,0,2.5)	9.99×10^7	5.59×10^7

Note: The coordinates of the center of mass position are relative to the tower top coordinate system.

concentrated masses and moments of inertia are applied on the two reference points to consider the influence of RNA. Each reference point coupled with the tower's top section to simulate their motion. The base of the model was fully fixed, constraining all displacements and rotations in three directions. Considering computational accuracy and efficiency, a mesh size of 0.25 m was selected for the FE model. Fig. 5 illustrates the FE model of hybrid tower.

The model first applied prestress in the static general analysis step, and then conducted frequency analysis in the linear perturbation analysis step. The NLGEOM parameter was employed in the analysis steps to ensure the final shape post prestressing serves as the base state for frequency analysis. Modal analysis was conducted using the Lanczos method within ABAQUS.

4.2. Validation

To verify the reliability of theoretical solutions for the hybrid tower's dynamic characteristics, the first three natural frequencies and vibration modes obtained from FE results were compared with theoretical calculations. The comparison of the first three natural frequencies is shown in Table 3, and the corresponding vibration modes are illustrated in Fig. 6. As seen, for the first three natural frequencies and vibration modes of the hybrid tower, the theoretical solutions agree well with the FE results, thereby validating the accuracy of the proposed method for analyzing the dynamic characteristics of hybrid towers based on adaptive vibration modes. The theoretical results for the first-order frequencies along the u and v directions closely agree with the FE results, with relative errors within 5 %, meeting engineering design requirements. Besides, due to the complexity of higher-order vibration modes, lower modes exhibit higher agreement than higher modes.

5. Discussion

5.1. Effect of different RNA simplified models on natural vibration frequency

To explore the effect of different simplified models of the RNA on the natural frequencies of hybrid tower, four commonly used RNA models

were employed: point mass (PM), eccentric point mass (EPM), eccentric point mass with moment of inertia (EPMJ), and eccentric double points mass with moment of inertia (EDPMJ), as depicted in Fig. 7.

Based on the FE calculation results, the relative errors between the natural frequencies calculated by the four simplified models and the FE calculation results are listed in Table 4. Since the MM model ignores the top RNA's asymmetry, the natural frequencies of the tower along both directions are the same. The PM and EPM models fail to consider the moment of inertia of the RNA, slightly overestimating the first natural frequency of the hybrid tower. For the second and third structural modes, relative deviations exceeding 10 % were observed in some results. Both the EPMJ and EDPMJ models comprehensively account for the effects of the RNA's mass eccentricity and moment of inertia, yielding results closer to each other. In summary, for the first bending mode of the tower, all four simplified models meet the precision requirements for engineering applications. However, for other bending modes, the RNA models that neglect moments of inertia show relative frequency deviations exceeding 10 %.

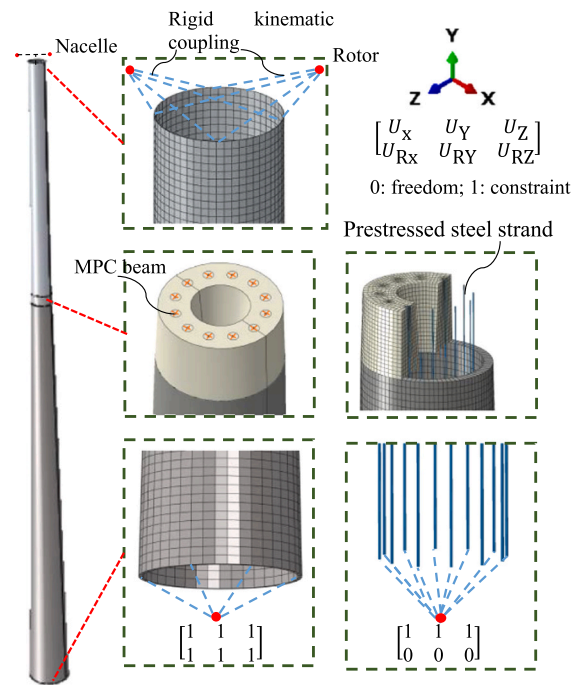
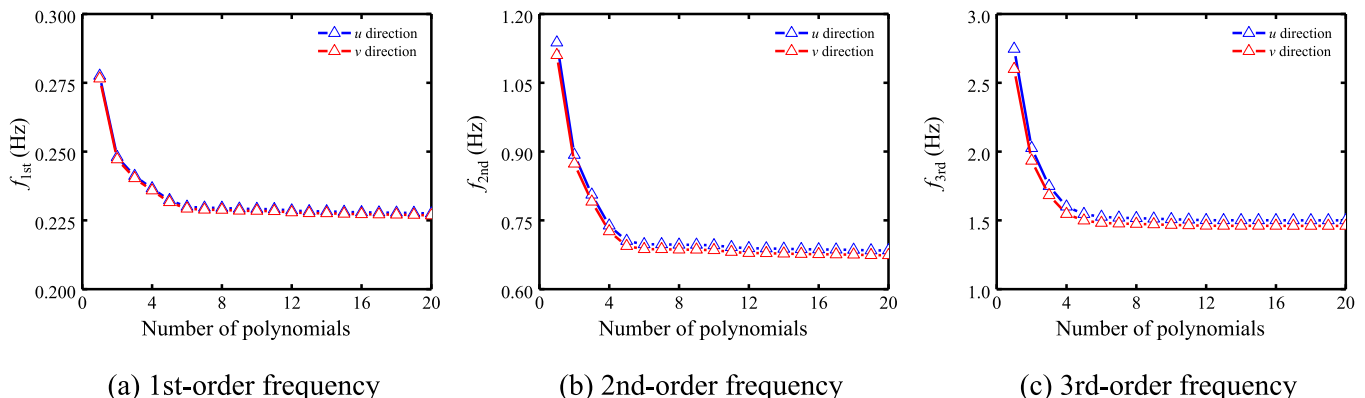
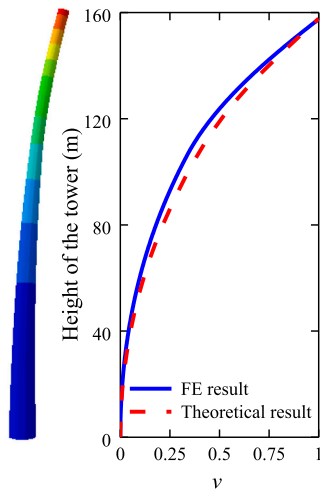
**Fig. 5.** FE model of hybrid tower.**Fig. 4.** Calculation results of the first three natural frequencies of hybrid tower.

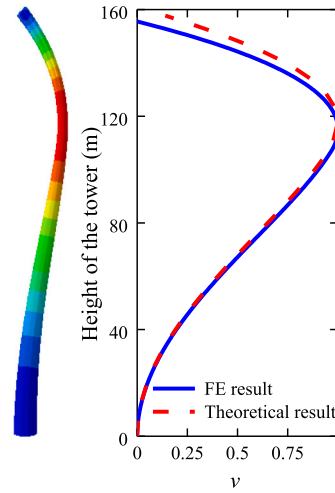
Table 3

Comparison of the first three natural frequencies between theoretical solutions and FE results.

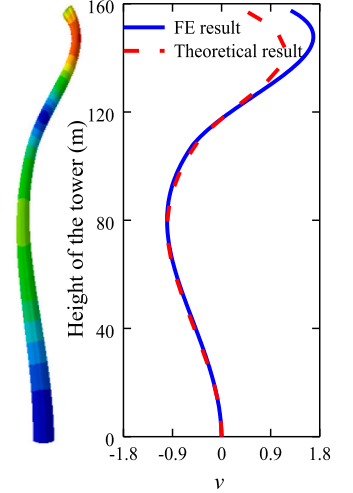
Direction	Order	FE results (Hz)	Theoretical solution (Hz)	Relative error (%)
v-direction	1st-order	0.2347	0.2292	-2.33
	2nd-order	0.6464	0.6871	6.30
	3rd-order	1.4306	1.4818	3.58
u-direction	1st-order	0.2389	0.2299	-3.73
	2nd-order	0.7033	0.6977	-0.80
	3rd-order	1.6636	1.5257	-8.29



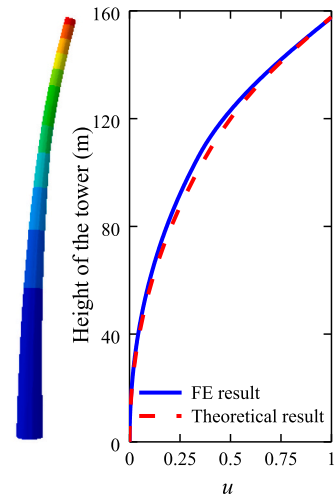
(a) 1st-order mode-v direction



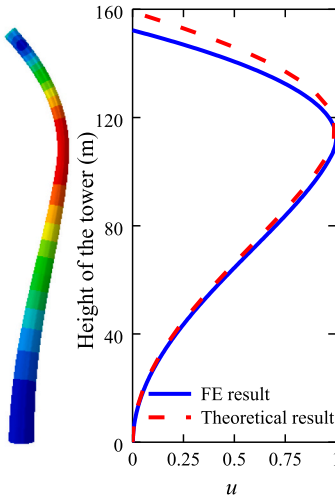
(b) 2nd-order mode-v direction



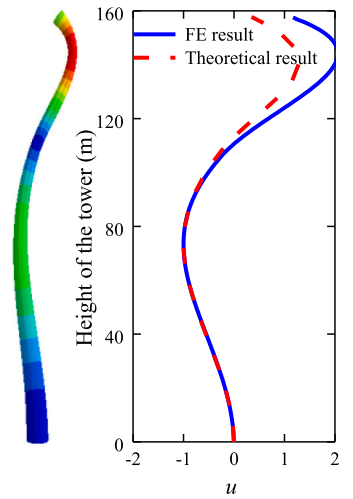
(c) 3rd-order mode-v direction



(d) 1st-order mode-u direction



(e) 2nd-order mode-u direction



(f) 3rd-order mode-u direction

Fig. 6. Comparison of the first three vibration modes between theoretical solutions and FE results.

5.2. Effect of different geometric dimensions on natural vibration frequency

The natural frequencies of the hybrid tower are influenced by a variety of features, including tower height, tower segment diameters, tower segment thicknesses, the proportion of concrete tower segment, and prestress level. To simplify the initial design process, it is essential to determine the significance of these different features and complete the optimization and selection of the most critical features. A machine learning model is employed to train a regression task and perform feature selection, identifying the most significant features impacting the

tower's natural frequencies. This aids in adjusting preliminary design schemes for the tower more effectively.

To eliminate multicollinearity issues among features, independent random sampling was conducted for all variables. The sampling ranges and intervals for each variable are detailed in Table 5. A dataset with a total sample size of 4096 was designed, and the first three natural frequencies for each sample were calculated using the theoretical method. The dataset was randomly shuffled and divided into training and testing sets in an 8:2 ratio. Gradient Boosting Decision Trees (GBDT), an efficient ensemble learning algorithm that enhances model prediction through serial training of multiple decision trees [28], was employed.

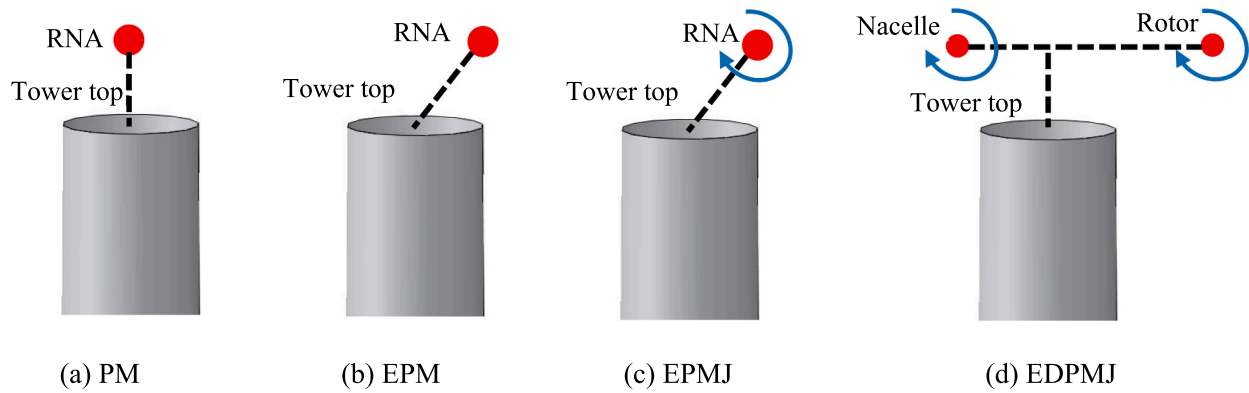


Fig. 7. Different RNA simplified models.

Table 4

Natural frequencies calculated by different RNA simplified models.

Direction	Order	PM (Hz)	Error (%)	EPM (Hz)	Error (%)	EPMJ (Hz)	Error (%)	EDPMJ (Hz)	Error (%)
v-direction	1st-order	0.2309	−1.96	0.2309	−1.96	0.2292	−2.33	0.2292	−2.33
	2nd-order	0.7133	10.34	0.713	10.30	0.6872	6.30	0.6871	6.30
	3rd-order	1.5922	11.30	1.5907	11.19	1.4820	3.59	1.4818	3.58
u-direction	1st-order	0.2309	−3.33	0.2309	−3.33	0.2300	−3.72	0.2299	−3.73
	2nd-order	0.7133	1.42	0.7133	1.42	0.6980	−0.75	0.6977	−0.80
	3rd-order	1.5922	−4.29	1.5922	−4.29	1.5270	−8.21	1.5257	−8.29

Table 5

Basic features and sampling range of the dataset.

Feature	Meaning of feature	Units	Sampling range	Sampling spacing
D_{cb}	Bottom diameter of concrete tower segment	m	[6,10]	0.05
D_{ct}	Top diameter of concrete tower segment	m	[3,5]	0.05
D_{sb}	Bottom diameter of steel tower segment	m	[3,5]	0.05
D_{st}	Top diameter of steel tower segment	m	[3,4]	0.05
t_{cb}	Bottom thickness of concrete tower segment	m	[0.2,0.4]	0.02
t_{ct}	Top thickness of concrete tower segment	m	[0.2,0.4]	0.02
t_{sb}	Bottom thickness of steel tower segment	m	[0.01,0.03]	0.002
t_{st}	Top thickness of steel tower segment	m	[0.01,0.03]	0.002
γ_{sc}	Concrete tower segment proportion	m	[0.1,0.9]	0.1
P	Prestress	MN	[0100]	10
H	Height of tower	m	[100,160]	5

The model was constructed and trained using the Scikit-learn machine learning framework in Python [29]. The main hyper parameters were set as follows: the number of decision trees was 100, the maximum tree depth (max_depth) was set to 2, and the learning rate was set to 0.1. As shown in Fig. 8, the training results indicate that the goodness of fit R^2 exceeded 0.98, demonstrating the model's excellent prediction accuracy for the natural frequencies of the hybrid tower. Thus, this model can be effectively used for feature importance analysis of the hybrid tower's natural frequencies.

In the trained GBDT model, feature importance is measured based on the average reduction in impurity brought by each feature. Fig. 9 illustrates the contribution of each feature to the model's predictions. Among the features, the total height of the tower and the proportion of the concrete segment are the most significant, with their importance substantially surpassing other features. The bottom diameter of the

concrete segment also significantly impacts the structural stiffness, ranking high in importance, while the diameters of other sections occupy moderate positions. However, the thickness of the tower sections has a limited effect on improving stiffness, resulting in a lower ranking in the feature importance analysis for natural frequencies. Additionally, the impact of prestress on the first natural frequency of the tower is notable and even exceeds the influence of the wall thickness of the tower segments.

To further investigate the specific impact of geometric dimensions on the natural frequencies of the hybrid tower, a parametric analysis was conducted on hybrid towers with heights of 100 m, 120 m, 140 m, and 160 m. Fig. 10 displays the results of the parametric analysis of the tower's natural frequencies. As illustrated in Fig. 10, the diameter and wall thickness parameters of the tower segments positively correlate with the natural frequencies of the tower, although the extent of the impact varies. The bottom diameter of the concrete segment has the most significant effect on the natural frequencies. When the bottom diameter of concrete segment increases from 6 m to 10 m, the first natural frequency of the towers with heights of 100 m, 120 m, 140 m, and 160 m increases by 80.1 %, 93.3 %, 116.0 %, and 161.7 %, respectively. The increase in the wall thickness of both the concrete and steel segments has a less pronounced effect on the natural frequencies, and increasing wall thickness does not significantly enhance stiffness. The height proportion of the concrete segment exhibits a non-linear relationship with the first natural frequency of the tower, initially increasing and then decreasing. This behavior is due to the combined effects of increased generalized stiffness and mass of the tower. This is primarily because an increase in the height proportion of the concrete tower segment enhances the generalized stiffness of the tower while also increasing its generalized mass. When the proportion of the concrete segment height is less than 0.7, the generalized stiffness of the tower dominates, leading to a continuous increase in the first natural frequency. The maximum first natural frequency is achieved when the height proportion is about 0.7. Beyond this proportion, the generalized mass of the tower dominates, causing a gradual decline in the first natural frequency. Additionally, an increase in prestress leads to a reduction in the generalized stiffness of the tower, thus decreasing the

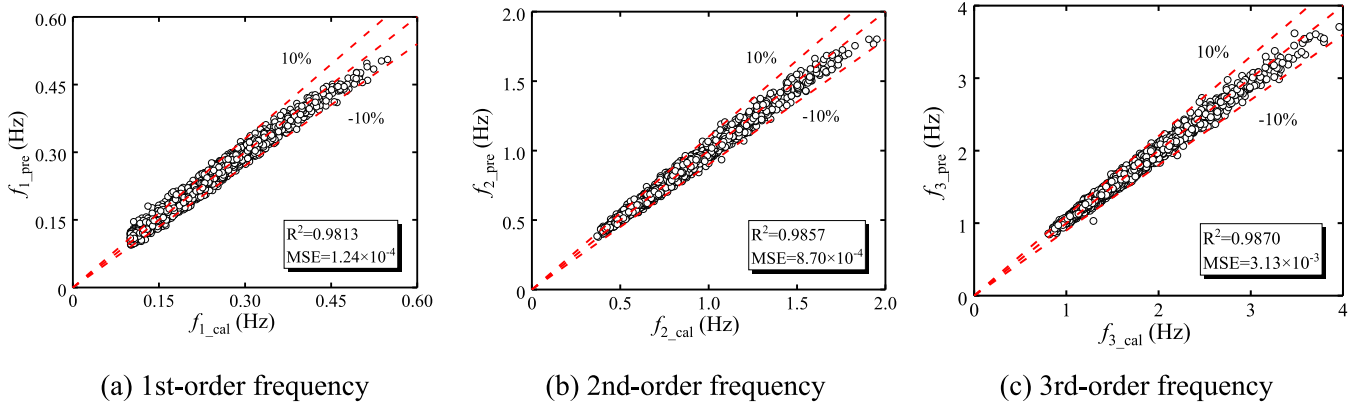


Fig. 8. Prediction performance of GBDT models. Note: f_{1_cal} , f_{2_cal} , and f_{3_cal} are the theoretical calculation results of the first-order, second-order, and third-order natural frequencies, respectively; f_{1_pre} , f_{2_pre} , and f_{3_pre} are the machine learning predicted results of the first-order, second-order, and third-order natural frequencies, respectively.

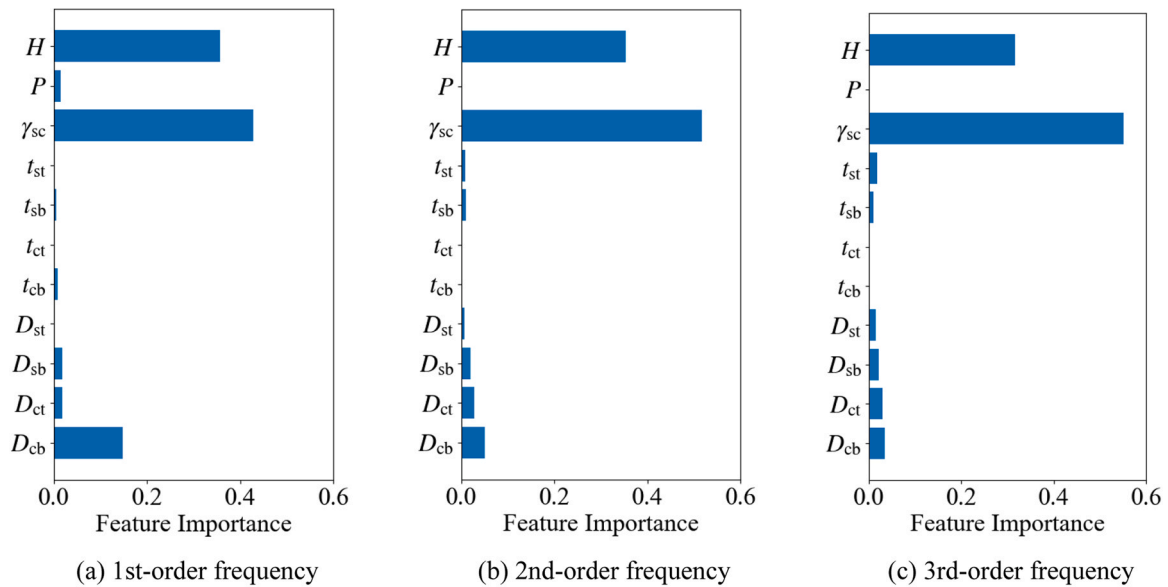


Fig. 9. Analysis of the feature importance of natural frequency.

natural frequency of the structure, particularly the first natural frequency.

6. Conclusion

An intelligent analysis of the dynamic characteristics is proposed for steel-concrete hybrid wind turbine tower, and the conclusion is summarized as follows:

- (1) The control equation of the hybrid tower was derived based on the virtual displacement principle and generalized coordinate method, considering the effects of mass eccentricity and moment of inertia of the RNA, variable cross-sections of the tower, and prestress. By employing the Rayleigh energy method and intelligent optimization algorithms, an adaptive vibration mode was constructed, and an efficient solution for the natural frequencies was achieved.
- (2) A refined finite element model using ABAQUS software was established to compute the natural frequencies and vibration modes of the hybrid tower. For the first three natural frequencies and vibration modes of the hybrid tower, the theoretical solutions

closely agree well with the FE results, with relative errors not exceeding 10 %.

- (3) The influence on the natural frequencies of four different simplified RNA models of point mass, eccentric point mass, eccentric point mass with moment of inertia, and eccentric double points mass with moment of inertia was explored. Results showed that all four simplified models met the engineering accuracy requirements for the first natural frequency of the hybrid tower, whereas models neglecting moments of inertia overestimated the frequencies for the second and third modes.
- (4) Machine learning models for predicting the first three natural frequencies of the hybrid tower were trained using GBDT, and the goodness of fit R^2 of the models exceeded 0.98, indicating excellent training efficacy. Feature importance analysis based on these models highlighted tower height, proportion of concrete tower segment height, and bottom diameter of concrete segment as significant factors influencing natural frequencies.
- (5) Parameter analysis was conducted on the natural frequencies of hybrid towers with hub heights of 100 m, 120 m, 140 m, and 160 m. Results indicated a strong positive correlation between the bottom diameter of concrete tower segment and natural frequencies. The proportion of concrete tower segment height

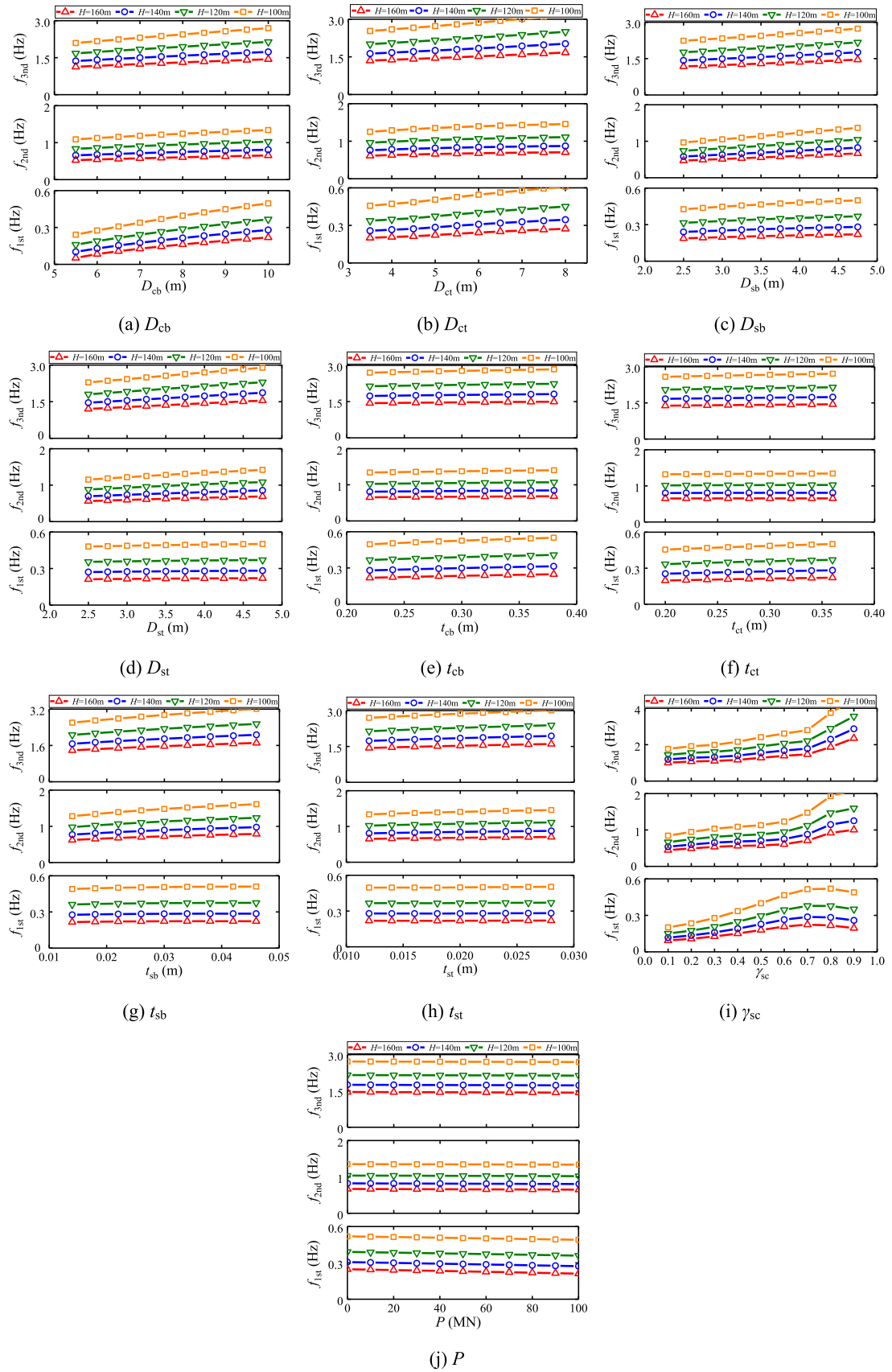


Fig. 10. Parametric analysis on natural frequencies.

showed an initial increase followed by a decreasing trend in the first natural frequency. Prestress exhibited a negative correlation with structural natural frequencies, while the increase in wall thickness of tower segments showed slight effects on natural frequency enhancements.

CRediT authorship contribution statement

Yuhang Wang: Validation, Supervision, Data curation. **Xuhong Zhou:** Validation, Supervision, Funding acquisition. **Liang Cao:** Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Jiepeng Liu:** Writing – review & editing, Supervision, Resources, Investigation, Conceptualization. **Yu Cheng:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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